Software Requirements Specification (SRS)

IEEE Std 830-1998 Compliant

Problem Statement #33: Micro-Lending & Peer Credit Risk Assessor

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1. Introduction & Purpose

This Software Requirements Specification defines the operational, algorithmic, architectural, and quality benchmarks for the Micro-Lending & Peer Credit Risk Assessor system. The platform bridges the retail credit gap by connecting unbanked and thin-file borrowers with retail peer lenders through deterministic heuristic underwriting and transparent reducing-balance amortization schedules.

2. Overall Product Perspective

The system operates under a 3-Tier Layered Architecture with decoupled service modules: (1) Loan & Borrower Management, (2) Credit Risk Evaluation Engine, (3) Amortization Engine, (4) P2P Marketplace & Escrow Service, (5) Payment & Settlement Gateway, and (6) Default Monitoring & Alert Service.

3. Key Functional Capabilities (FR-001 to FR-005)

• **FR-001 (Credit Risk Assessment):** Calculates Debt-to-Income (DTI), employment stability index, and historical repayment consistency to assign credit scores (300-850) and risk tiers (Low, Medium, High).

• **FR-002 (Amortization Schedules):** Produces exact reducing-balance monthly EMI schedules ensuring mathematical balance conservation ($\sum$ Principal = Loan Amount).

• **FR-003 (P2P Escrow Marketplace):** Allows peer lenders to review listings and commit fractional funds into escrow with strict overfunding prevention.

• **FR-004 (Repayment & Dividends):** Settles borrower repayments and disburses proportional dividends to lenders atomically within ACID transactions.

• **FR-005 (Default Alerting):** Automatically issues T+3 day grace period reminders and T+15 day formal default warnings with credit score downgrades.

4. Non-Functional Benchmarks

• **NFR-001:** 200 concurrent transactions sustained with 99th-percentile latency < 350 ms and zero ledger drift.

• **NFR-002:** Field-level AES-256 encryption at rest, TLS 1.3 in transit, and continuous WAL backup ensuring RPO ≤ 15 mins and RTO ≤ 1 hour.